School of Computing

RESEARCH UNIVERSITY

ANSWER BOOKLET

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MATRIC NO.	A19EC0065

COURSE CODE & TITLE	SECR1213 NETWORK COMMUNICATION
GROUP / SECTION	06
LECTURER'S NAME	DR AHMAD FARIZ
EXAMINATION DATE	16 FEBRUARY 2021
EXAMINATION VENUE	ONLINE E-LEARNING

QUESTION	FOR EXAMINER'S USE ONLY
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TOTAL	

INSTRUCTIONS TO CANDIDATES:

1. *Fill in the above details clearly.*
2. *Write your answers at the space provided for each question.*
3. *Convert this answer booklet into pdf formatted file.*
4. *Submit the pdf file on the provided link in the elearning.*

QUESTION 1

(a)(i)

192.168.4.50

192.168.4.51

192.168.4.52

192.168.4.53

192.168.4.54

192.168.4.55

192.168.4.56

192.168.4.57

192.168.4.58

192.168.4.59

192.168.4.60

192.168.4.61

192.168.4.62

192.168.4.63

192.168.4.64

192.168.4.65

(a)(ii)

i) $4 = 2^2$

Subnet 0 : 11000000.10101000.00000100.00000000

Subnet 1 : 11000000.10101000.00000100.01000000

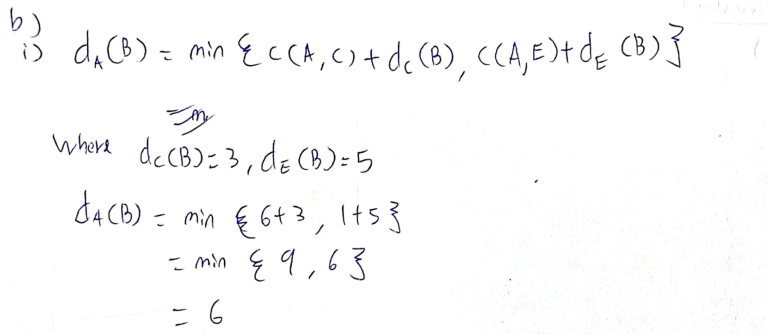
Network address for subnet 1 : 192.168.4.64

Broadcast address for subnet 1 : 192.168.4.127

Range of usable address : 192.168.4.65 – 192.168.4.126

Subnet mask : 255.255.255.192 or 11111111.11111111.11111111.11000000

(b)(i)



(b)(ii)

Step	N'	D(C),P(C)	D(E),P(E)	D(D),P(D)	D(B),P(B)
0	A	6,A	1,A	∞	∞
1	AE	3.E	-	6,E	∞
2	AEC	-	-	5,C	11,C
3	AECD	-	-	-	6,D
4	AECDB	-	-	-	-

(c)

Fragment 1	Length=1860	ID=629	Offset=0	Flag=1
Fragment 2	Length=1860	ID=629	Offset=233	Flag=1
Fragment 3	Length=1860	ID=629	Offset=465	Flag=1
Fragment 4	Length=1480	ID=629	Offset=698	Flag=0

QUESTION 2

(a)(i)

Collision can still occur because the distance and proprogration delay cause the nodes to be unable to hear each other transmission in time.

(a)(ii)

The difference between random access and taking turn protocols is the occurrence of collisions in random access protocol, while there is collison occurs in taking-turn protocols.

(b)(i)

Ip address, mac address and TTL

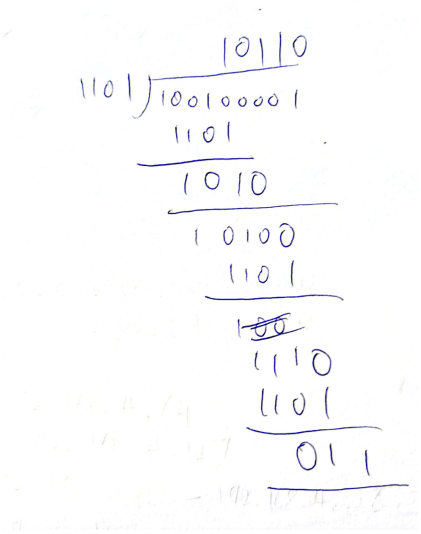
(b)(ii)

Ip address	MAC address
192.168.1.2	AA-AA-AA-AA-AA-AA
192.168.1.4	CC-CC-CC-CC-CC-CC

192.168.1.5	01-01-01-01-01-01
192.168.2.3	01-01-01-01-01-01

(c)

Since the remainder is not zero, thus there is error for the received data.



QUESTION 3

(a)(i)

For hidden terminal problem, B and A can hear each other and B and C can hear each other as well.But, A and C cannot hear each other and this means that A and C does not aware of their interference at B.

For fading problem, B and A can hear each other and B and C can hear each other as well. But , A and C cannot hear each other and this implies that there is an interference at B.

(a)(ii) IEEE 802.11

(b)

In the 3G network, users firstly connect the radio access networks to the public networks. After that, the core networks that interoperates with the MSC. The core networks are made up of two components which are SCGNs and GGSNs. SCGNs is responsible for delivering datagrams to or from the mobile nodes in the radio access networks to which it is attached. SCGNs provides user authorization and handoff, maintaing location information about the active mobile nodes and also forward datagram between mobile nodes and GGSN.GGSN acts as a gateway that connects multiple SGSN into the larger network.